

Charting the Path to Bifröst: The 2026 ESIT Data-to-Information Competition on Robust CSI-Based Localization and Machine Unlearning

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with contributions from the finalist teams:

1st Place – KUDFI:	Esraa Fahmy Othman, Yu Tian, and Hang Zou
2nd Place – Yggdrasil Embeddings:	Hanjun Park, Rakhmania Amalia, Batool Memoona, and Hanan Al-Tous
3rd Place – Asgard Autoencoders:	Mohamed El Mehdi Makhoulouf, Oussama Harrak, and Loukas Duque
4th Place – Solo Skald:	Michele Banfi

Abstract—The 2026 European School of Information Theory (ESIT) Data-to-Information (D2I) Competition, titled “*Charting the Path to Bifröst*”, invited students to tackle wireless localization from real-world Channel State Information (CSI) under *imperfect data*. Departing from earlier D2I editions, the challenge placed *data reliability* at its centre: participants had to localize a transmitter from distributed CSI when both measurements and position labels are corrupted, and then to *unlearn* the influence of unreliable contributors from an already-trained model. The competition was built around the DICHASUS distributed channel-sounder dataset and unfolded in two graded tasks—*Learning under Noisy Conditions* (robust CSI-based localization) and *Unlearning and Data Reliability* (machine unlearning evaluated by a membership-inference attack)—preceded by an ungraded warm-up. This article reports the competition’s motivation and format, describes the dataset, tasks, and evaluation metrics, reports participation, and presents the approaches of the finalist teams.

I. INTRODUCTION

The *Data-to-Information* (D2I) Competition is an initiative of the IEEE Information Theory Society Student and Outreach Subcommittee, created to engage students with problems at the intersection of information theory and machine learning. Its guiding idea is to chart the passage from raw *data*, which this days is abundant, unstructured, and sometimes noisy, to actionable *information*, where the principles of information theory give shape and meaning to complexity. Through hands-on, data-driven challenges, the competition aims to spark students’ curiosity toward these connections.

Previous editions established the format. The first, *DeepVerse 6G* at ISIT 2023 [1], addressed CSI compression at

a user equipment reporting to a base station with multi-modal side information. The second, the 2024 *Bits and Bots Competition* [2], cast participants as both bot designer and bot detector, highlighting hypothesis testing and statistical learning for cybersecurity. The third, *Wireless Wanderlust* at ISIT 2025 [3], introduced CSI-based localization through *channel charting* [4] and distributed inference under communication constraints. The ISIT 2026 edition, *The Still Mirror*, addressed learning, optimization, and security in reconfigurable intelligent surface (RIS)-assisted wireless systems.¹

The ESIT 2026 edition, presented in this article, is the *first* D2I Competition held in conjunction with the *European School of Information Theory* (ESIT), which took place in Nordfjordeid, Norway. Under the theme “*Charting the Path to Bifröst*”, it continues the localization trajectory of *Wireless Wanderlust* but pivots to a question of growing practical importance: how to learn reliably when the data itself cannot be trusted. In deployed wireless systems, both CSI measurements and the position labels used to supervise localization are imperfect—sensors drift, phases decalibrate, antennas drop out, and a fraction of contributors report corrupted trajectories. The competition therefore asked participants first to build localization models that stay accurate and reliable under such corruption (*robust learning* [5]), and then to surgically remove the influence of unreliable contributors from a trained model without retraining it from scratch (*machine unlearning* [6], [7]).

¹<https://data-to-information.github.io/isit2026/>

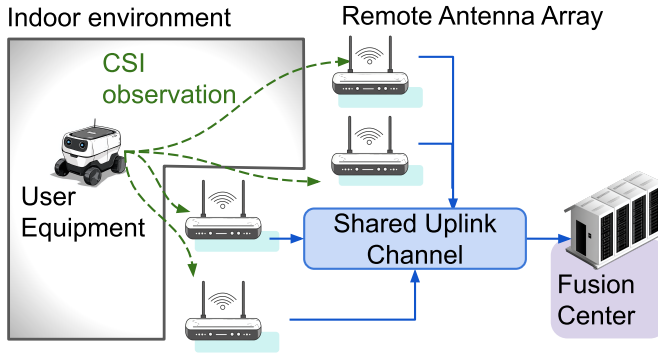


Fig. 1. Distributed CSI-based localization setup underlying the competition. A mobile user equipment (UE) transmits pilot signals whose Channel State Information (CSI) is observed by $R = 4$ distributed remote antenna arrays. The arrays forward their observations over a shared uplink channel to a fusion center, which estimates the UE position. In the competition, the CSI observations and the training and validation position labels released to participants are deliberately corrupted (Sec. I-A).

The theme draws on the Norse legend of Bifröst, the shimmering bridge connecting Midgard to Asgard. Where the ancients needed divine intervention to cross, today’s explorer needs only a stable cellular connection: given noisy, partial, and imperfect measurements of the wireless channel, can one still chart a path across the bridge? This framing motivated a friendly, thematically consistent set of Viking-inspired team names and a narrative arc from foundations to advanced reliability challenges.

The dataset underpinning the competition is drawn from the *DICHASUS* (Distributed CHannel Sounder by University of Stuttgart) collection [8], measured at the Arena2036 research campus. *DICHASUS* combines a distributed, massive-MIMO measurement infrastructure with over-the-air synchronization to deliver phase-, time-, and frequency-coherent CSI across widely spaced antenna arrays, tagged with high-precision ground-truth positions. Building on this dataset, and deliberately corrupting it, the competition invited student teams to design learning algorithms that are robust to label noise, feature corruption, and faulty data contributions.

The competition was introduced at ESIT 2026 and hosted on two Kaggle competitions (one per graded task) with a shared Slack community, and culminated in a live online session on July 7, 2026, in which the selected teams presented their approaches; the winners were announced on July 8, 2026.

The competition’s final live event was generously sponsored by the ELLIS Society, Madrid Unit, with additional support from Comunidad de Madrid and the Universidad Carlos III de Madrid.

The competition website—canonical source for both learning and unlearning tasks—can be found online.²

A. Competition Setup

The competition abstracts a realistic indoor/industrial localization scenario, illustrated in Fig. 1. A single-antenna mobile

transmitter, denoted as user equipment (UE), transmits pilot signals that are received by $R = 4$ distributed remote antenna arrays, each of which records CSI in a set of frequency bins. The arrays forward their observations over a shared uplink channel to a fusion center, which must infer the transmitter’s position from the aggregated CSI. For the robust learning task (named Task 1 below), the public release comprised labeled training and validation recordings and an unlabeled test recording.

What distinguishes the ESIT 2026 edition is that the released data are *deliberately corrupted*. Released position labels are transformed and further perturbed by temporal drift and jitter, while CSI measurements are altered through gain and phase offsets, additive noise, and compressed-frequency-bin dropout. Participants therefore cannot assume clean supervision; the challenge is to balance average accuracy against reliability, avoiding catastrophic localization failures. In the second half of the competition, a further complication is introduced: a subset of users contributes systematically corrupted trajectories, and their influence must be removed from a pretrained model after their identities are partially revealed.

B. Task Overview

The competition was organized as a sequence of tasks of increasing difficulty. An ungraded warm-up (Task 0) introduced the data and metrics; the two graded tasks (Task 1 and Task 2) formed the two halves of the challenge, hosted as separate Kaggle competitions. The final leaderboard was obtained combining performance across both graded tasks.

- **Task 0: Foundations (warm-up, ungraded).** A guided introduction to CSI-based localization: how CSI tensors are represented, how position labels and timestamps are used, how to train simple localization baselines, how to compute the evaluation metrics, and how temporal structure can be exploited. This task was released as an annotated notebook with helper modules.
- **Task 1: Learning under Noisy Conditions (graded).** This is the main robust-localization challenge. CSI measurements and observed positions carry realistic artifacts such as calibration mismatch, measurement noise, compressed-frequency-bin dropout, and smooth trajectory drift. Participants train models that predict organizer-held 2D reference positions (x, y) for every test sample and remain accurate and reliable despite the corruption. Temporal models may exploit the sequence structure of the data (see Section II).
- **Task 2: Unlearning and Data Reliability (graded).** An advanced challenge on machine unlearning. Participants start from a *fixed* pretrained model whose training data mixed reliable (*retain*) and unreliable (*forget*) contributors. Once the identities of a subset of forget-set samples are revealed, participants must remove their influence through fine-tuning (or other means) under a strict budget, so that the model treats forgotten samples as if they were unseen.

Notation. We write vectors in bold lowercase (e.g., $\mathbf{p}$), and tensors in bold uppercase (e.g., $\mathbf{H}$). We denote the Euclidean

²See <https://data-to-information.github.io/esit2026/>.

norm by $\|\cdot\|_2$, and an estimate of $\mathbf{p}$ by $\hat{\mathbf{p}}$. Estimated and reference 2D positions are compared through the per-sample error $e_i = \|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2$.

II. PROBLEM SETTING

A. Dataset Description

The competition data are derived from the DICHASUS distributed channel sounder [8], measured at the Arena2036 research campus. The environment is served by $R = 4$ distributed remote antenna arrays (Fig. 1). In Task 1’s release, each CSI observation is a complex-valued tensor

$$\mathbf{H} \in \mathbb{C}^{4 \times 2 \times 4 \times 64},$$

corresponding to the four receive antenna arrays, a 2×4 antenna layout per array, and 64 compressed frequency bins per antenna element. The target is an organizer-defined 2D reference position $\mathbf{p} \in \mathbb{R}^2$. Training and validation labels are released as (x, y, z) , where x and y are the horizontal position coordinates and z is the vertical coordinate in the released coordinate system. Test reference positions are held by the organizers, and evaluation uses only (x, y) .

Each sample carries metadata: a unique `id`, a `sequence_id`, and a `relative_time` within the sequence. Because measurements are organized into temporally ordered sequences, temporal models (RNNs, transformers, state-space models, smoothing) may process each `sequence_id` independently, sorting rows by `relative_time` and resetting state between sequences. Task 1’s development data consist of labeled training and validation splits; only the test split is unlabeled. To preserve the integrity of the challenge, competitors were prohibited from using the raw, uncorrupted DICHASUS measurements or any metadata derived from them beyond the official release.

B. Task 1: Robust Localization Metric

Localization quality is measured by a single lower-is-better score that jointly rewards *accuracy* and *reliability*. For a set of n test samples with per-sample errors $e_i = \|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2$, define the *Mean Euclidean Distance Error* (MEDE) and the *90th-percentile radius* (R90):

$$\text{MEDE} = \frac{1}{n} \sum_{i=1}^n e_i, \quad (1)$$

$$\text{R90} = \text{percentile}_{90}(\{e_i\}_{i=1}^n). \quad (2)$$

The competition score is their equally weighted sum,

$$\mathcal{L} = \frac{1}{2} \text{MEDE} + \frac{1}{2} \text{R90}. \quad (3)$$

MEDE captures average accuracy, while R90 penalizes the tail of large errors. Minimizing (3) thus discourages catastrophic localization failures in addition to improving typical accuracy. Submissions are made through Kaggle as a CSV of predicted (x, y) per test `id`. The public leaderboard uses $\sim 30\%$ of the test set and the private leaderboard the remaining $\sim 70\%$, which determines the final task ranking.

A simple Ridge-regression pipeline was released as Task 1’s baseline, together with starter code for loading, preprocessing, local evaluation, and submission formatting.

C. Task 2: Machine Unlearning

Task 2 studies *machine unlearning* [6], [7], whose main purpose is removing the influence of specific training observations from an already-trained model. The motivation is data reliability—a subset of users contributes corrupted trajectories due to faulty GPS or sensor drift, and their influence should be removed if it degrades performance, ideally without the cost of retraining from scratch.

In this task, participants receive a training set of real-valued CSI tensors of shape $\mathbf{H} \in \mathbb{R}^{2 \times 32 \times 64}$, where the three axes index the real and imaginary components, the $32 = 4 \times 2 \times 4$ receive antenna elements across the four arrays, and the 64 compressed frequency bins, respectively, together with positions and a `is_forget` flag partitioning the data into a *retain* set (62.5%, `is_forget` = 0) and a *forget* set (37.5%, the noisier observations, `is_forget` = 1). A pretrained convolutional position predictor is provided as a fixed checkpoint, and it was trained on the mixture of retain and forget data (100 epochs, learning rate 10^{-3} , MSE loss, batch size 64). Participants had to start from this exact checkpoint and apply an unlearning algorithm under a strict budget: at most 50 fine-tuning epochs at learning rate 10^{-4} , any objective function, variable batch size. Training a new model from scratch is not permitted.

Success was measured by a Membership Inference Attack (MIA). The intuition is that if the forget set has truly been unlearned, the model should treat those samples like unseen data, producing errors that are *distinguishable* from those on the retain set. Concretely, per-sample localization errors e_i are computed on the training and test sets, where a fixed logistic-regression classifier (the default `sklearn` implementation provided in the starter notebook) is fit to predict `is_forget` from the scalar error. Its accuracy on the held-out test set is the leaderboard score. *Higher accuracy is better*, since it indicates that forgotten samples have become separable from retained ones. For reference, the released baseline (gentle fine-tuning on the retain set only) scores ≈ 0.61 , while a model retrained from scratch on the retain set alone reaches ≈ 0.84 , which can be seen as a useful target for what successful unlearning under the budget can approach.

D. Overall Ranking

The two graded tasks were ranked independently on their private Kaggle leaderboards, and the overall standing combined the two. For each eligible team, let R_1 and R_2 be its private-leaderboard ranks on Task 1 and Task 2. The overall score is the rank sum $R_1 + R_2$ (lower is better), with ties broken by the average normalized relative difference to the best team on each task.

III. COMPETITION OUTCOMES

A. Participation and Prizes

The D2I-ESIT 2026 competition drew ten registered teams (twenty participants in total) spanning a range of institutions and experience levels, from solo entrants to teams of four. Team formation was finalized on-site after the in-person introduction to the competition at ESIT 2026 on June 2, with remote participants assisted by the organizers.

On the basis of combined performance across the two graded tasks, four teams were selected to present their solutions at the competition’s live session on July 7, 2026. This closing session was generously sponsored by the ELLIS Society, Madrid Unit, which provided speaker awards recognizing the invited teams’ presentations and their contributions to the event. The awards, totaling €1,000, were allocated to the four presenting teams as €500, €250, €150, and €100 (the last is a Junior Prize), following the overall competition standings announced on July 8, 2026:

- 1) **KUDFI** (1st place);
- 2) **Yggdrasil Embeddings** (2nd place);
- 3) **Asgard Autoencoders** (3rd place);
- 4) **Solo Skald** (4th place).

B. Finalist Approaches

The following team reports summarize the methodologies of the four finalist teams.

1) *First Place: KUDFI: Team: Esraa Fahmy Othman, Yu Tian, and Hang Zou* (Digital Future Institute at Khalifa University, Abu Dhabi, UAE).

Overview. KUDFI treated the two tasks as separate error-shaping problems: Task 1 reduced localization tail error, while Task 2 made the fixed MIA scorer separate retain and forget rows through prediction error. The combined entry ranked first overall, with small Task 1 models and a simple Task 2 fine-tune rather than external data or pretrained weights.

Data representation. The released Task 1 CSI tensor was converted into two inputs. One branch used a (2, 32, 64) log-magnitude plus delay-domain magnitude image, with per-sample standardization and an IFFT across subcarriers; phase was discarded there because it was corrupted. The second branch used (8, 36, 5) intra-array conjugate products of energy-normalized delay-tap responses. The per-array phase noise cancels in the product, leaving intra-array relative phase. Compressed-frequency-bin dropout was not explicitly imputed, masked, or marked. Dropped bins entered the IFFT, where their effect spread across delay taps rather than concentrating. Temporal sequence structure was not used, and the released train/validation splits were kept as given.

Task 1 – robust localization. KUDFI ensembled two fusion models over the same two branches: xcv, a 3.87 M parameter cross-attention fusion regressor, and fcv, a 5.24 M parameter sequence-style fusion regressor with cross-attention refinement followed by joint convolution. Each architecture used six seeds. Training used denoising auto-encoder pre-training for the Branch A encoder, which addressed the feature corruption

implicitly, plus C-Mixup with $\sigma = 0.8$, AdamW, cosine scheduling, and 120 epochs. The loss was smooth-L1 plus 0.05 times the mean of the worst 10% of per-sample Euclidean errors in the batch. The final equal blend scored 1.08086 private and 1.09230 public on 0.5(MEDE+R90).

Task 2 – machine unlearning. KUDFI fine-tuned the provided checkpoint with one MSE objective: retain rows kept their given position as the target, while every forget row target was shifted by $M = 16$ in the same global [1, 0] direction. This made the model express corruption detection as a position offset, which the fixed logistic MIA scorer could read through the scalar prediction error. The selected submission scored 0.97748 private and 0.99144 public, against about 0.61 for naive fine-tuning and about 0.84 for from-scratch retraining; a shorter 40-epoch run under the same settings performed comparably.

Takeaways. The largest Task 1 gain came from matching the loss to the metric tail through the worst-10% CVaR term. The two Task 1 architectures made different errors on hard samples, so equal averaging improved the public and private scores. In Task 2, the shared forget-target offset was the load-bearing choice.

2) *Second Place: Yggdrasil Embeddings: Team: Hanjun Park* (POSTECH, Pohang, South Korea); Rakhmania Amalia, Batool Memoona, and Hanan Al-Tous (Aalto University, Espoo, Finland).

Overview. Yggdrasil Embeddings first isolated the dominant bottleneck in each task and then targeted it directly. For Task 1, the team combined a transformer-based localization model [9] with channel-informed techniques. Task 2 is evaluated using a fixed logistic-regression model applied to the model’s positioning error, so the team formulated unlearning as the deliberate shaping of that error. Channel charting [4], building on prior work by team members [9], [10], provided the common foundation for both tasks.

Data representation. For Task 1, the team represented the complex-valued CSI through amplitude and cosine/sine phase features. For Task 2, the team used a standardized stack of the real and imaginary components. Corrupted frequency bands, represented by zeroed rows, are restored using mask-gated residual reconstruction, while clean rows remain unchanged. The team retained the released train/validation/test splits and recovered the temporal structure from `sequence_id` and `relative_time`.

Task 1 – robust localization. The team used a transformer operating on 32 channel tokens [9], [10], one per antenna element, with each token represented by a 320-dimensional vector formed from 5 feature planes across 64 frequency bins. The team trained the model with a score-aligned objective combining a distance-Huber term and an R_{90} term, directly reflecting the competition metric $\mathcal{L}$. The distance-Huber term applies the Huber penalty to the per-sample Euclidean local-

ization error $r_i = \|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2$:

$$L_\delta(r_i) = \begin{cases} \frac{1}{2}r_i^2, & r_i \leq \delta, \\ \delta(r_i - \frac{1}{2}\delta), & r_i > \delta. \end{cases} \quad (4)$$

Robustness is further improved through phase-rotation invariance derived from the channel structure. By the submission deadline, these components reduced the private Kaggle leaderboard score from 1.26735 to 1.10166.³

Task 2 – machine unlearning. The team fine-tuned the CNN so that its positioning error alone discriminated between corrupted and clean rows. For the forget samples, the team denoised the target positions toward estimates obtained from channel charting [9], causing the evaluated positioning error to reflect the injected corruption. The team then used corruption signals extracted from the released data to construct a detector and encoded its output into the positioning error through transductive fine-tuning, further sharpening the separation. The final membership labels are produced by the mandated logistic regression, achieving `is_forget` accuracies of 0.99842/0.99633 (private/public).

Takeaways. The team emphasized deriving *invariances* directly from the channel rather than treating them as tunable heuristics, and interpreting a ceiling on any individual signal as a limitation of that signal—not of the task itself.

3) *Third Place: Asgard Autoencoders: Team:* Mohamed El Mehdi Makhoulf, Oussama Harrak, and Loukas Duque (Inria, Lyon, France).

Overview. Asgard Autoencoders used a representation-learning strategy for Task 1 and a rank-based reliability strategy for Task 2. The guiding principle was to preserve the useful structure of the distributed CSI, learn robust features before supervised localization, and avoid late corrections that overfit fragile validation or public-leaderboard signals.

Data representation. For Task 1, the CSI tensor $\mathbf{H} \in \mathbb{C}^{4 \times 2 \times 4 \times 64}$ was kept as four receiver-array tokens. For each array, the team concatenated log-magnitude, real and imaginary parts, frequency phase differences, unwrapped phase, normalized magnitude, and cross-array phase, giving $7 \times 512 = 3584$ features per array. The cross-array phase was computed elementwise as $\arg(\mathbf{H}_a \odot \mathbf{H}_{a-1}^*)$, with $\arg$ denoting complex phase. Features were standardized and clipped. The encoder was a compact array-token transformer, inspired by recent transformer-based CSI positioning models [11]: a shared token MLP mapped each array vector to a 128-dimensional token, array-identity embeddings identified the four receiver arrays, and two transformer blocks with four attention heads mixed the array tokens.

Task 1 – robust localization. The localization model was trained in three steps. First, a BYOL-style self-supervised encoder [12] was pretrained from short-range temporal positive pairs: two CSI samples close in time were encouraged

to have similar representations after independent augmentations such as noise, array dropout, frequency masking, and phase jitter. Second, this encoder was refined with temporal and masking objectives. Same-sample consistency made two augmented views of the same CSI agree; nearby-sample prediction encouraged the current CSI representation to predict a representation one to four time steps ahead; and masked-view consistency forced the encoder to recover stable information when parts of the CSI were hidden. Finally, a localization head was trained with a freeze-unfreeze schedule and a Huber-based supervised loss. The selected Task 1 submission was a small blend: $\hat{\mathbf{p}} = 0.90\hat{\mathbf{p}}_{\text{temporal}} + 0.10\hat{\mathbf{p}}_{\text{linear}}$, where $\hat{\mathbf{p}}_{\text{temporal}}$ came from the temporally refined SSL encoder and $\hat{\mathbf{p}}_{\text{linear}}$ came from the first-stage SSL encoder with a simpler linear head and standardized coordinates. This submission scored 1.14350 public and 1.23231 private. Post-hoc diagnostics showed that the simpler linear/std model was actually best privately (1.14419 private, 1.16087 public), while graph smoothing and residual corrections generally hurt robustness.

Task 2 – machine unlearning. Task 2 was treated as reliability ranking. A first score was obtained from robust position/CSI deviation features: position deviations, radius deviations, and per-subcarrier CSI amplitude profiles were normalized with median/MAD statistics and passed to logistic regression. A second score came from gradient-ascent fine-tuning (GA+FT) teachers initialized from the organizer model: retain samples were fine-tuned normally, while forget samples received a small ascent step to amplify their error signature. Since the two scores had different scales, both were converted to percentile ranks and blended as $r_i = 0.90r_i^z + 0.10r_i^{\text{teacher}}$. The top 50% were labeled forget-like. This rule scored 0.94502 private (0.93765 public); the z-score-only variant already reached 0.92787 public, showing that the robust deviation signal carried most of the gain.

Takeaways. For Task 1, preserving array structure and using SSL before noisy supervised fine-tuning helped, but late complex corrections overfit. For Task 2, robust ranks were safer than raw scores, and GA+FT was useful only as a small auxiliary signal.

4) *Fourth Place: Solo Skald: Team:* Michele Banfi (Politecnico di Milano, Milan, Italy), competing solo.

Overview. Solo Skald used architectural diversity to limit overfitting to the small set of labeled trajectories. The entry combined four model families for Task 1 and two complementary unlearning methods for Task 2, using equal-weight averaging for localization and averaging the two methods’ error signals for unlearning.

Data representation. Only CSI magnitude $|\mathbf{H}|$ was retained, since phase-based experiments degraded performance under the released corruption. A shared backbone processed each receive array, and its outputs were flattened before position regression to preserve frequency structure. Samples were ordered by `relative_time` within each `sequence_id` for temporal modeling and smoothing. The channel-charting objective also used the unlabeled test CSI.

³Continued refinement of the same pipeline reached 0.99955 shortly after Task 1 deadline. This result is included for completeness, but it was submitted too late to count toward the official standings, which are based on the deadline score of 1.10166.

TABLE I
FINALIST RESULTS SUMMARY. TASK 1 REPORTS THE PRIVATE-LEADERBOARD LOCALIZATION SCORE $\frac{1}{2}(\text{MEDE}+\text{R90})$ (LOWER IS BETTER); TASK 2 REPORTS THE PRIVATE-LEADERBOARD MEMBERSHIP-INFERENCE ACCURACY (HIGHER IS BETTER); OVERALL IS THE COMBINED RANK (SEC. II).

Team	Task 1 $\frac{1}{2}(\text{MEDE}+\text{R90})\downarrow$	Task 2 MIA acc. $\uparrow$	Overall rank
KUDFI	1.08086	0.97748	1
Yggdrasil Embeddings	1.10166	0.99842	2
Asgard Autoencoders	1.23231	0.94502	3
Solo Skald	1.34047	0.85654	4

Task 1 – robust localization. The final ensemble comprised 18 models: six magnitude-based convolutional neural networks (CNNs), four temporal models with bidirectional gated recurrent units (GRUs), four echo state networks, and four channel-charting CNNs. The magnitude CNNs used a Huber loss. The GRU models processed windows of 21 consecutive frames and predicted the center position. The echo state networks used fixed recurrent reservoirs and ridge-regression readouts, providing a distinct source of errors. The charting CNNs combined a supervised Huber loss with a triplet loss that brought temporally nearby samples together in the embedding and separated feature-mined negatives. Equal-weight averaging scored 1.34047 private and 1.35218 public. Learned combiners overfit the labeled trajectory, while adding distinct model families improved the ensemble.

Task 2 – machine unlearning. Internal activations of the pretrained CNN already distinguished retain and forget samples, but parameter dampening degraded both sets. The team therefore combined relabeling with activation-trajectory divergence. Relabeling replaced each forget target with the mean position of its ten nearest retain neighbors in CSI-magnitude feature space after principal component analysis, then fine-tuned the model. Better position estimates made the error against corrupted labels reflect the injected corruption. Activation-trajectory divergence used the frozen baseline as a teacher: retain activations after rectification were anchored to the teacher, while forget activations before rectification were pushed away using a hinge-cosine loss with a norm penalty to prevent collapse. This method also used the corrected forget targets. Averaging the two methods’ error signals scored 0.85654 private and 0.92787 public, compared with approximately 0.61 for the baseline and 0.84 for retraining from scratch.

Takeaways. Magnitude features and architectural diversity supported robust localization with limited labeled data. For Task 2, correcting unreliable position labels and modifying internal activations provided complementary error signals for the fixed membership-inference scorer.

C. Cross-Team Summary

Across the four finalists, several themes recur. For Task 1, every team concluded that corrupted phase was better suppressed than modeled—through magnitude-only features (Solo

Skald), intra-array conjugate products (KUDFI), structured per-array features (Asgard), or an explicit phase-rotation invariance (Yggdrasil)—and that matching the training loss to the metric’s tail (a worst-10% CVaR term, a distance-Huber + R_{90} objective, or a plain Huber loss) mattered more than raw model capacity. Architectural diversity combined by equal-weight averaging was repeatedly the safest route to a lower R90, whereas learned combiners tended to overfit the single labeled trajectory. For Task 2, the decisive realization, shared by the strongest entries, was that the fixed logistic membership-inference scorer reads only the scalar position error, so unlearning is best posed as *shaping* that error: KUDFI shifted every forget target by a common offset, Yggdrasil and Solo Skald denoised forget labels so that the residual equals the injected corruption, and Asgard ranked reliability from robust deviation statistics. Because the contamination lived in the position labels rather than the model parameters, the methods that recovered the corruption magnitude transferred cleanly from training to test and approached (or, for the top two teams, even exceeded) the retrain-from-scratch reference.

IV. CONCLUSION

The ESIT 2026 edition of the Data-to-Information Competition, “*Charting the Path to Bifröst*”, brought the D2I series to the European School of Information Theory for the first time and shifted its focus toward the reliability of data-driven wireless inference. By corrupting both measurements and labels, and by requiring the surgical removal of unreliable contributions from a trained model, the competition confronted students with two increasingly relevant facets of trustworthy machine learning—robustness and unlearning—grounded in a real distributed-CSI dataset. The finalist solutions ranged from carefully regularized, architecturally diverse ensembles that suppressed corrupted phase and matched the training loss to the metric’s tail, to unlearning schemes that reframed a fixed membership-inference scorer as an error-shaping objective and recovered the injected corruption from the labels themselves, illustrating the creativity and technical depth of the participants and reaffirming the value of hands-on challenges at the intersection of information theory and machine learning. Future editions could test whether these approaches generalize to new trajectories and corruption patterns, and assess unlearning through both retained localization accuracy and agreement with models retrained on reliable data. Such evaluations would help distinguish gains tailored to a fixed scorer from more general improvements in data reliability.

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Fig. 2. The logo of the the ELLIS Society.

competition’s final live event. ELLIS (European Laboratory for Learning and Intelligent Systems) supports fundamental research in machine learning and artificial intelligence across Europe, connecting scientific advances with societal and economic impact. It also promotes collaboration and training for young researchers. The society logo is depicted in Fig. 2, their website is in the footnote ⁴. Finally, we thank all participants for their enthusiasm, creativity, and rigor throughout the D2I-ESIT 2026 competition.

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